# Project 3:-Maven Selenium Project – Automating Product Selection Using Selenium

## Step 1: Create a Maven Project Using Terminal

mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenSeleniumApp01 -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false

```
harith@harith-VirtualBox:~/devops/test$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenSeleniumApp01 -DarchetypeArtifactId=maven
-archetype-quickstart -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< org.apache.maven:standalone-pom >-------------------
[INFO] Building Maven Stub Project (No POM) 1
[INFO] --------------------------------[ pom ]---------------------------------
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] ----------------------------------------------------------------------------
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO] ----------------------------------------------------------------------------
[INFO] Parameter: basedir, Value: /home/harith/devops/test
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenSeleniumApp01
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/harith/devops/test/MyMavenSeleniumApp01
[INFO] ----------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ----------------------------------------------------------------------------
[INFO] Total time:  1.913 s
[INFO] Finished at: 2026-03-17T20:56:48+05:30
[INFO] ----------------------------------------------------------------------------
```

## Step 2: Install ChromeDriver

google-chrome --version

sudo apt install jq
sudo apt install curl

wget https://storage.googleapis.com/chrome-for-testing-public/$LATEST_DRIVER/linux64/chromedriver-linux64.zip

unzip chromedriver-linux64.zip
sudo mv chromedriver-linux64/chromedriver /usr/local/bin/chromedriver
sudo chmod +x /usr/local/bin/chromedriver
chromedriver –version

## Step 3: Add Selenium Dependency in pom.xml

```
<dependency>
<groupId>org.seleniumhq.selenium</groupId>
<artifactId>selenium-java</artifactId>
<version>4.29.0</version>
</dependency>
```

**Step 4: Configure Maven Build Plugins**

Add the following plugins in the **pom.xml** file.

```
<build>
<plugins>

<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.8.1</version>

<configuration>
<source>1.8</source>
<target>1.8</target>
</configuration>

</plugin
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-jar-plugin</artifactId>
<version>3.0.1</version>

<configuration>
<archive>
<manifestEntries>
<Main-Class>com.example.App</Main-Class>
</manifestEntries>
```

</archive>

</configuration>

</plugin>

</plugins>

</build>

Open

**pom.xml**
~/devops/test/MyMavenSeleniumApp01

```xml
<build>
  <plugins>

    <!-- Compiler -->
    <plugin>
      <groupId>org.apache.maven.plugins</groupId>
      <artifactId>maven-compiler-plugin</artifactId>
      <version>3.8.1</version>
      <configuration>
        <source>11</source>
        <target>11</target>
      </configuration>
    </plugin>

    <!-- Jar -->
    <plugin>
      <groupId>org.apache.maven.plugins</groupId>
      <artifactId>maven-jar-plugin</artifactId>
      <version>3.0.1</version>
      <configuration>
        <archive>
          <manifest>
            <mainClass>com.example.App</mainClass>
          </manifest>
        </archive>
      </configuration>
    </plugin>

  </plugins>
</build>
```

**Step 5: Inspect Elements in the Website**

Steps:

1. Open the website
   https://automationexercise.com/products
2. Inspect the page for elements required, sucha s add-to-cart button.

**Step 6: Create App.java**

Java Code:

```
package com.example;

import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeDriver;
public class App {
    public static void main(String[] args) throws InterruptedException {
        WebDriver driver = new ChromeDriver();
        // Open Products Page
        driver.get("https://automationexercise.com/products");
        // Maximize Browser Window
        driver.manage().window().maximize();
        // Add Product 4 to Cart
        driver.findElement(By.cssSelector("[data-product-id='4']")).click();
        Thread.sleep(2000);
        driver.findElement(By.cssSelector(".btn.btn-success.close-modal.btn-block")).click();

        // Add Product 5 to Cart
        driver.findElement(By.cssSelector("[data-product-id='5']")).click();
        Thread.sleep(2000);
        driver.findElement(By.cssSelector(".btn.btn-success.close-modal.btn-block")).click();

        // Add Product 6 to Cart
        driver.findElement(By.cssSelector("[data-product-id='6']")).click();
        Thread.sleep(2000);
        driver.findElement(By.cssSelector(".btn.btn-success.close-modal.btn-block")).click();

        // Open Cart Page
        driver.get("https://automationexercise.com/view_cart");

        // Refresh Cart Page
        driver.navigate().refresh();

        // Wait for 10 seconds
```

```
        Thread.sleep(10000);

        // Close Browser
        driver.quit();
    }
}
```

**Step 7: Build the Maven Project**

Run the following command:

mvn clean install

```
harith@harith-VirtualBox:~/devops/test/MyMavenSeleniumApp01$ mvn compile
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< com.example:MyMavenSeleniumApp01 >------------------
[INFO] Building MyMavenSeleniumApp01 1.0-SNAPSHOT
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MyMavenSeleniumApp01 ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/harith/devops/test/MyMavenSeleniumApp01/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:compile (default-compile) @ MyMavenSeleniumApp01 ---
[INFO] Nothing to compile - all classes are up to date
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  0.733 s
[INFO] Finished at: 2026-03-17T21:08:14+05:30
[INFO] ------------------------------------------------------------------------
harith@harith-VirtualBox:~/devops/test/MyMavenSeleniumApp01$
```

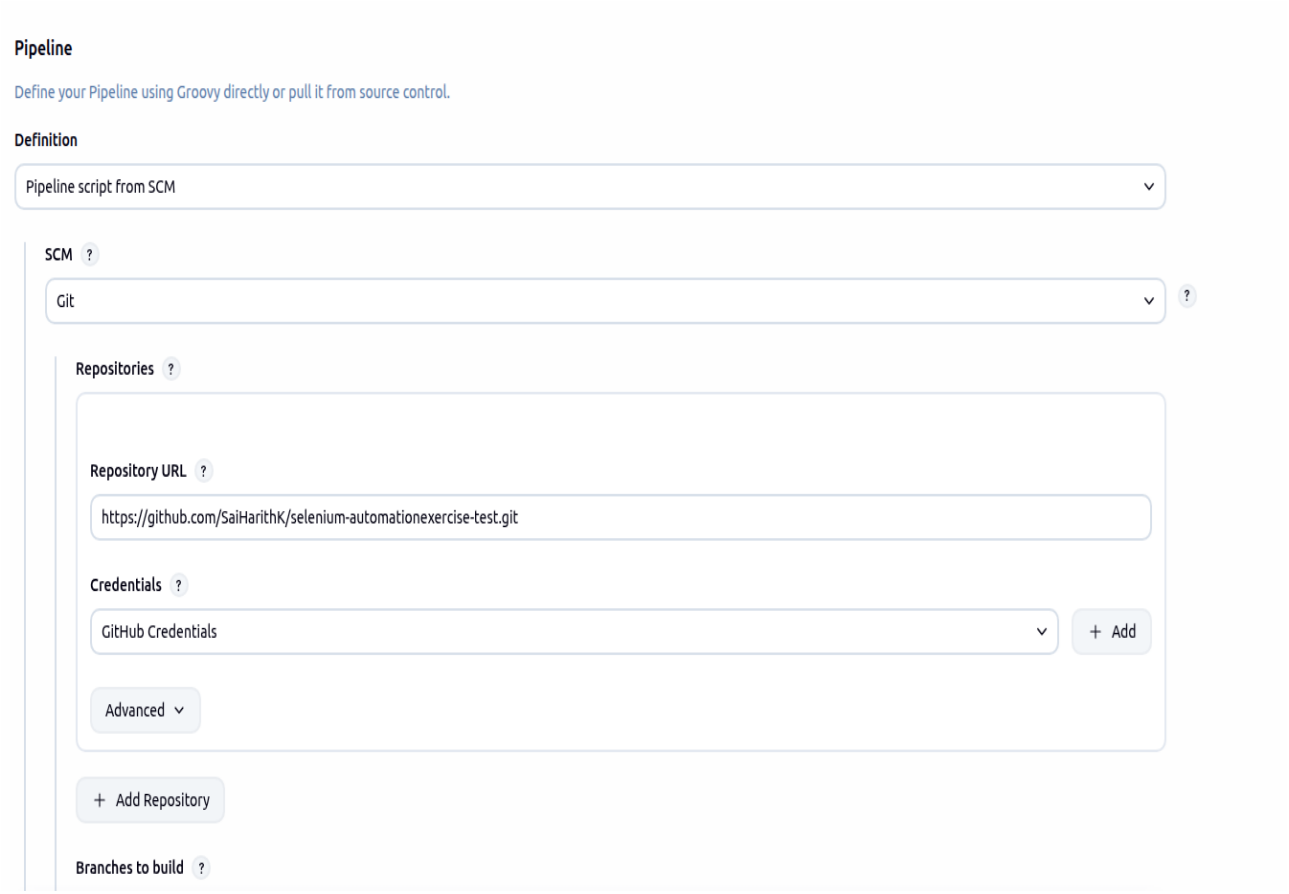
mvn exec:java -Dexec.mainClass=”com.example.App”

```
harith@harith-VirtualBox:~/devops/test/MyMavenSeleniumApp01$ mvn exec:java -Dexec.mainClass="com.example.App"
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< com.example:MyMavenSeleniumApp01 >------------------
[INFO] Building MyMavenSeleniumApp01 1.0-SNAPSHOT
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- exec-maven-plugin:3.1.0:java (default-cli) @ MyMavenSeleniumApp01 ---
```

**Step 8:** Pipeline

-login to jenkins account

-create a new item and configure the following settings




**Step 9:** Edit Jenkins file

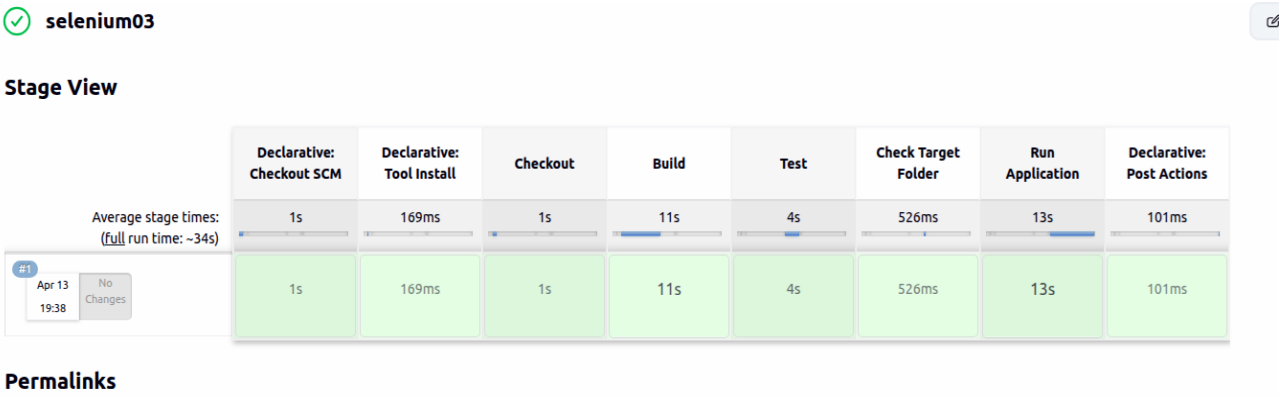
-Create a Jenkinsfile and add the following code:

```
pipeline {
   agent any  // Use any available agent
   tools {
      maven 'Maven'  // Ensure this matches the name configured in Jenkins
   }
   stages {
      stage('Checkout') {
         steps {
            git branch: 'main', url: 'https://github.com/SaiHarithK/selenium-automationexercise-test.git'
         }
      }

      stage('Build') {
```

```
            steps {
                sh 'mvn clean package'  // Run Maven build
            }
        }
        stage('Test') {
            steps {
                sh 'mvn test'  // Run unit tests
            }
        }

stage('Check Target Folder') {
    steps {
        sh 'ls -l target'
    }
}
stage('Run Application') {
    steps {
        sh 'java -jar target/AutomationExerciseTest-1.0-SNAPSHOT.jar'
    }
}
}
  post {
        success {
            echo 'Build and deployment successful!'
        }
        failure {
            echo 'Build failed!'
        }
    }
}
```

**Step 10:** Build

Click on build now and check if its successfull